

Credit Fraud Analysis

Dataset Overview

- Total Records: 31281
- Total Features: 10
- Duplicate Records Found: 0
- Missing Values Found: 0

Business Problem

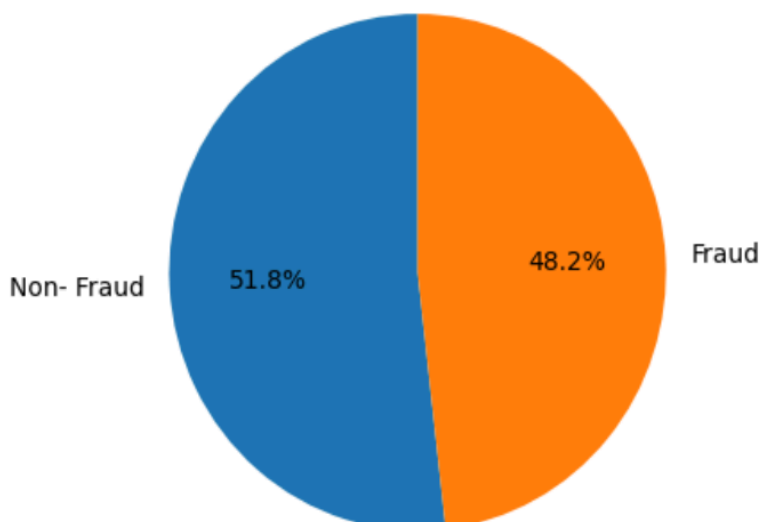
In recent years, Financial Institutions are facing challenges to detecting fraudulent transactions in real time. Fraud can lead to money loss, customer dissatisfaction and security risk. The project aims to analyze the fraudulent transactions pattern and build a system that predicts fraudulent activity.

Research Questions

- Does previous fraud history increase future fraud probability?
- Are foreign transactions riskier?
- Does merchant risk score affect fraud occurrence?
- Does transaction amount influence fraud?
- Is the risk score related to transaction amount?
- What is the sign of a fraudster?

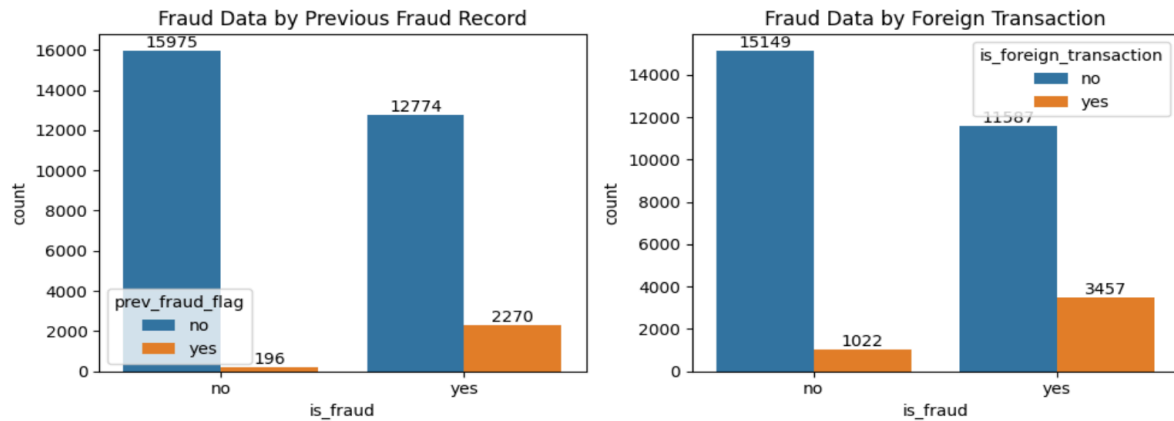
Analyze and Findings

Distribution of Fraudulent Transactions



The accompanying pie chart shows the percentage of transactions that are fraudulent and those that are not. There are approximately 48.2% identified as fraudulent transactions which have a significant impact on the company's business.

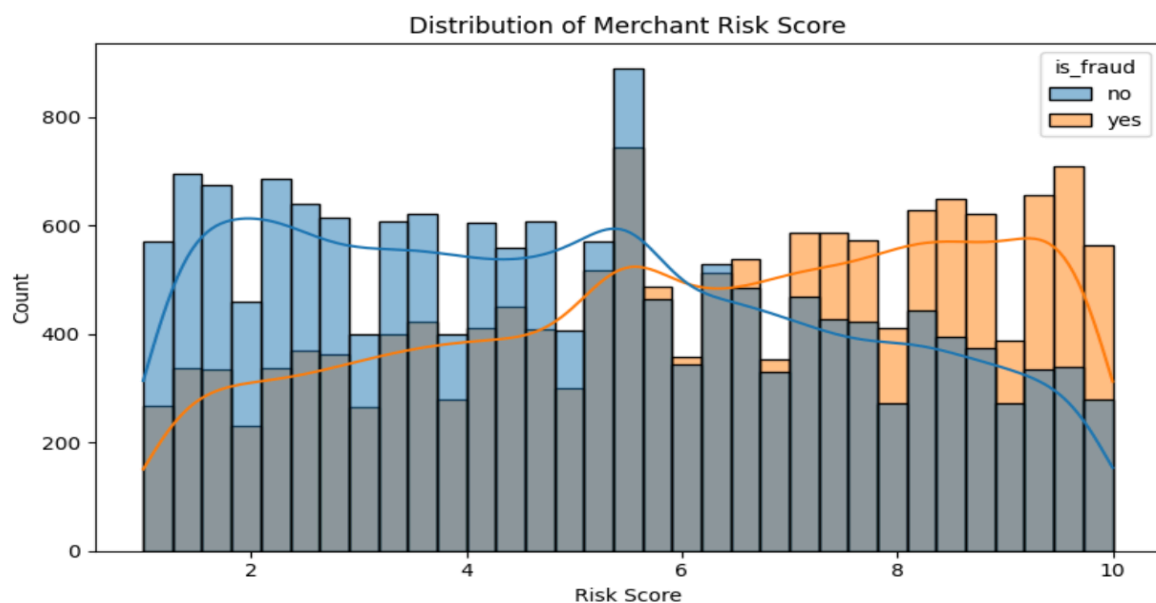
Fraud Detection by Different Features



The analysis indicates that users with prior fraud records are significantly more likely to commit fraud (15%) compared to users with no previous fraud history (1%).

Around 23% of fraud transactions were foreign transactions. This indicates that foreign transactions have a noticeable association with fraud and may require additional security monitoring.

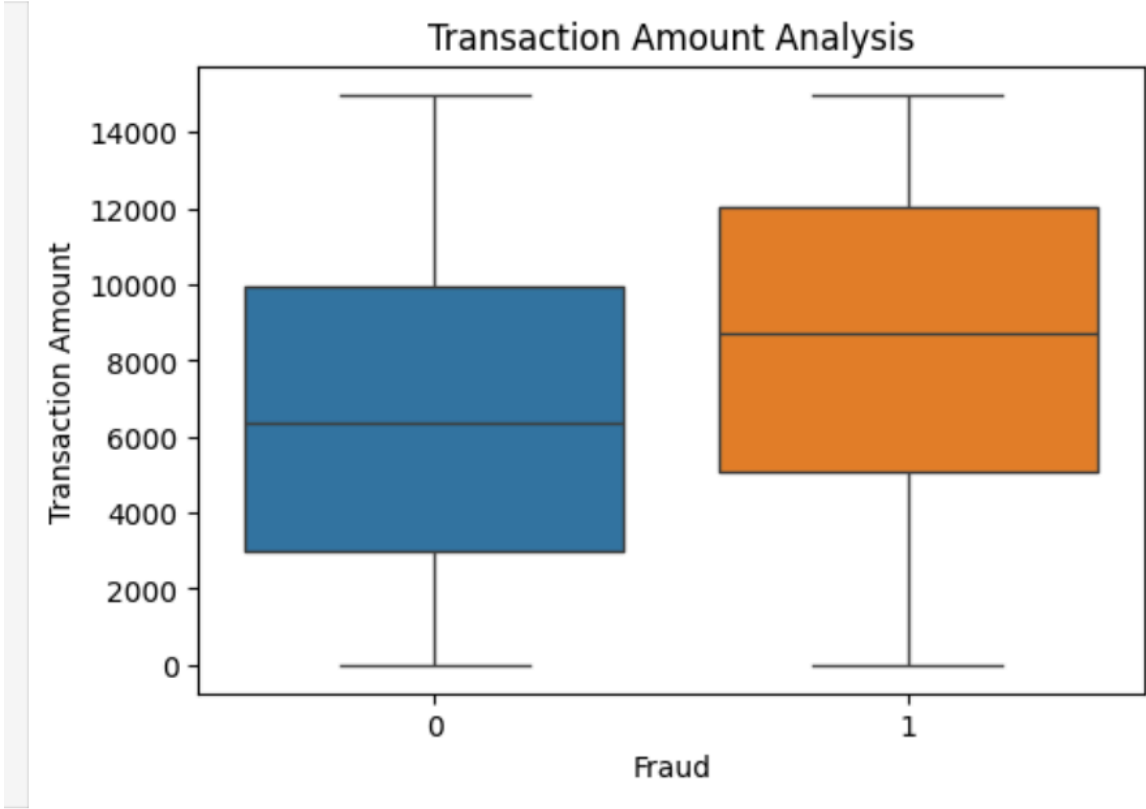
Risk Score Analysis



The histogram graph above shows that transactions associated with high-risk

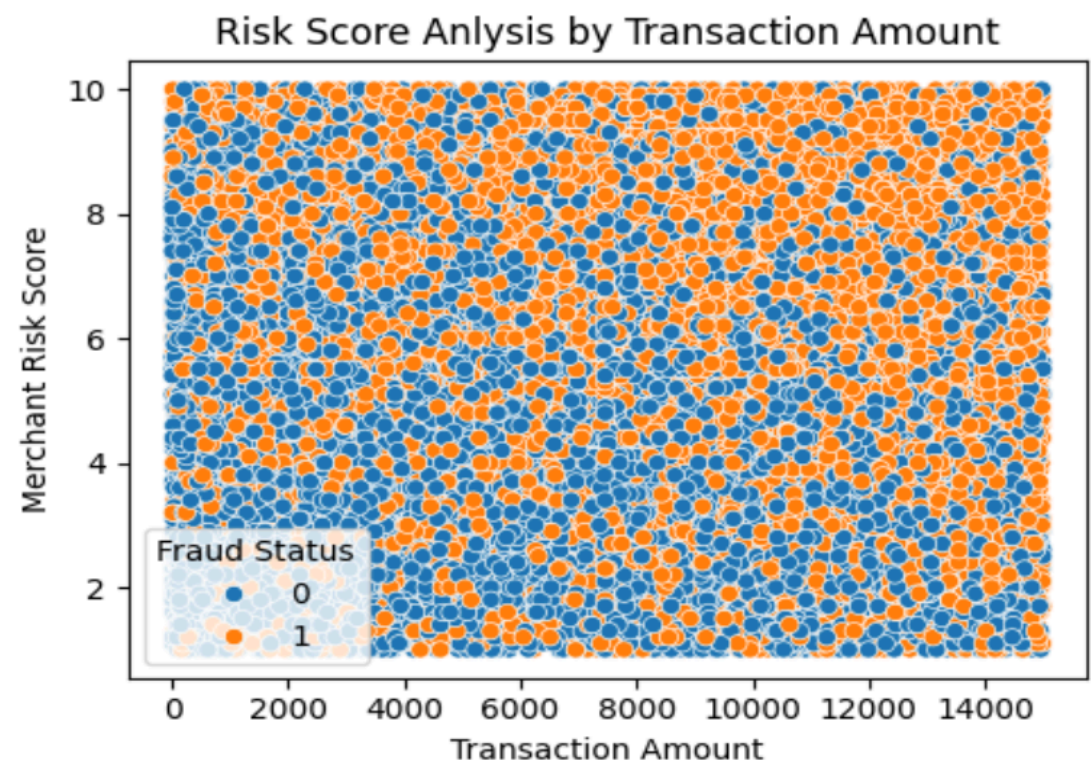
merchants demonstrated a significantly greater probability of fraud compared to the merchant who has a low-risk score.

Transaction Amount Analysis



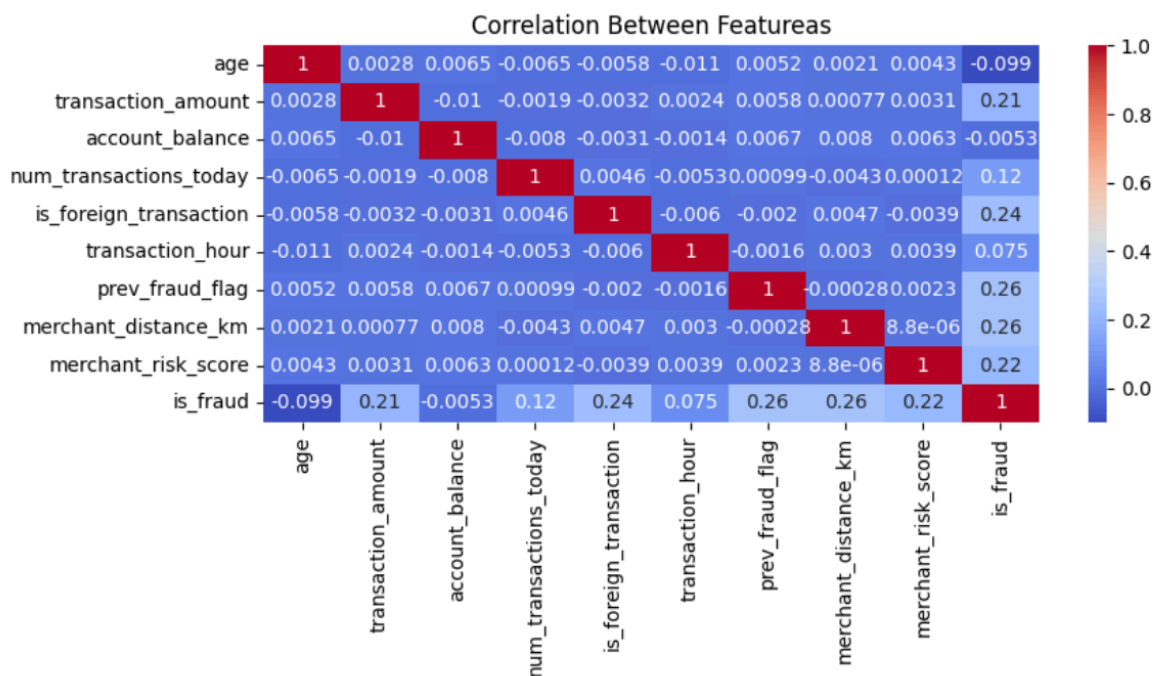
The boxplot graph demonstrates that fraudulent median transaction amount is 8500, where non-fraudulent median transaction amounts generally have higher amounts than legitimate ones.

Risk Score Analysis by Transaction Amount



We have made a scatter plot graph to analyze the relation between transaction amount and risk score. As we can see, when the risk score increases along with the transaction amount.

What Makes a Transaction Riskier?



The graph shows that the fraud is mainly connected to transaction behavior rather than customer personal information. High-value transactions, foreign payments, and transactions made from distant locations have a higher risk of fraud.

Suggestions

1. Fraud is often repetitive; the company should therefore maintain a “high monitoring customer list” for customers who have previous fraud involvement.
2. Foreign transactions have a higher probability of fraud. Fraudsters may exploit international payment systems because they are harder to verify and track quickly. So companies require location-based verification and add extra security to foreign transactions.
3. Fraud cases increase significantly when the risk score is high. Whenever a transaction comes from a high-risk score merchant, the system can request additional verification, temporarily hold the transaction or send for manual review.

4. Fraud transactions have a higher amount compared to normal transactions. The company can reduce the fraud by requiring biometric confirmation or triggering instant fraud alerts for unusually large payments.
5. A single factor does not cause fraud; fraud occurs when multiple risky behaviors happen together. The company should invest in real-time AI or machine learning fraud detection systems that continuously learn customer behavior patterns and avoid fraud.